

Supplementary Material for “Before the First Token: Auditing Affective Modality Conflict at the Pre-generation Interface”

APPENDIX A DATA AND CODE AVAILABILITY

Code and derived artifacts will be made available to reviewers upon request routed through the submission portal. Upon acceptance, we will release the implementation, trained readout weights, and evaluation scripts publicly.

The datasets used in this work (CMU-MOSI, CMU-MOSEI, CH-SIMS v2) are publicly available from their respective sources.

APPENDIX B PER-BACKBONE CALIBRATION SUMMARY

Table I summarizes the calibrated parameters for each backbone model under the corresponding protocol.

APPENDIX C ADDITIONAL EXPERIMENTAL RESULTS

A. Layer Selection Across Models

Fig. 1 reports layer-wise R^2 for all evaluated backbones. The optimal layer ℓ^* is selected as the layer maximizing average R^2 across modality conditions on the calibration holdout.

B. State Space Visualization

Fig. 2 visualizes the calibrated Affective State Space ($\mathcal{D}$ - $\mathcal{R}$ plane) for each model. The dashed lines represent the calibrated thresholds τ (vertical) and δ (horizontal). The logistic regression curve (dark blue line) illustrates the arbitration bias transition from one modality to the other as the split significance $\mathcal{D}$ increases.

C. Readout Performance under Conflict

Fig. 3 demonstrates the degradation of linear readout performance (R^2) as the semantic conflict between modalities increases. While unimodal probes (e.g., Text-only or Audio-only) often remain robust, the joint-modality readout (Purple bars) suffers in high-conflict scenarios (L3-L4), confirming the existence of internal interference.

D. Taxonomy Distribution

Fig. 4 reports the distribution of the four taxonomy states (Dominant, Balanced, Consensus, Confusion) across different conflict levels. As the conflict level increases (L1 to L4), we observe a shift in the distribution, with the proportion of Dominant and Balanced states varying by model architecture.

APPENDIX D NON-LINEAR READOUT ABLATION

A. MLP Architecture

The non-linear readout uses a two-layer MLP:

$$f_{\text{MLP}}(\mathbf{h}) = \mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2 \quad (1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{256 \times d_h}$ and $\mathbf{W}_2 \in \mathbb{R}^{2 \times 256}$.

B. Layer-wise R^2 Comparison

Fig. 5 compares layer-wise R^2 between linear and MLP readouts. The MLP readout shows monotonic improvement toward the final layer, suggesting exploitation of non-affective cues that destabilizes the geometric interpretation.

C. Impact on Conflict Metrics

Table II, III, and IV compare $\mathcal{D}$ and $\mathcal{R}$ distributions under linear vs. MLP readout on the $L1 - L4$ domain.

APPENDIX E COMPUTATIONAL OVERHEAD DETAILS

A. Benchmark Configuration

- **Hardware:** Single NVIDIA A100-80GB GPU
- **Inference engine:** vLLM v0.12.0 with parallel prefill
- **Batch size:** 1 (single-sample latency measurement)

B. Latency and Throughput Analysis

We evaluate the computational efficiency of the proposed SplitBrain architecture using the vLLM engine. Fig. 6 illustrates the latency metrics across different prompt templates.

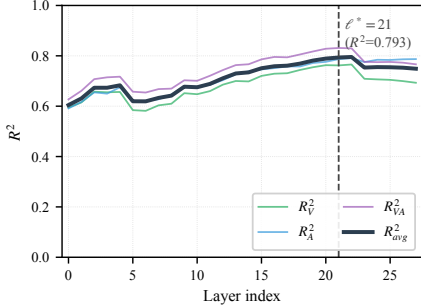
Table V summarizes the statistical performance. We report three key metrics:

- **TTFT (Parallel):** Time to First Token under parallel execution (Stage I).
- **Serial Output Time:** Total generation time for a single request processed sequentially.
- **Parallel Full Load:** Time to complete the entire batch in parallel.

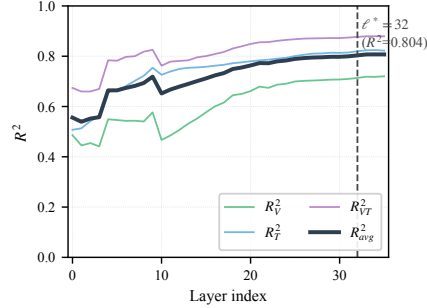
Responsiveness: As shown in Fig. 6, the Parallel TTFT (Blue line) remains highly stable ($\sigma = 0.120$) regardless of template complexity. Notably, the average time to start generating the first token in parallel mode (4.22 s) is faster than the average time required to fully complete a single serial task (5.84 s). This indicates that the system provides immediate

TABLE I
PER-BACKBONE CALIBRATION SUMMARY. ALL THRESHOLDS ARE CALIBRATED ON THE CONSISTENCY DOMAIN $\mathcal{B}$ USING THE 95TH PERCENTILE RULE AND FROZEN FOR EVALUATION.

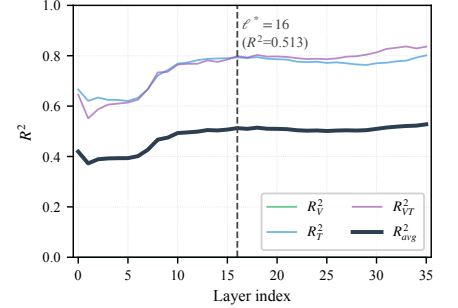
Backbone	Protocol	K	ℓ^*	κ	τ	δ	λ	ϵ
Qwen2.5-Omni-7B	Omni (V, A)	5	21	0.2186	0.3470	0.5974	0.01	10^{-6}
Qwen3-VL-8B	VL (V, T)	5	32	0.2179	0.5679	0.5888	0.01	10^{-6}
InternVL3_5-8B	VL (V, T)	5	16	0.2973	0.9318	0.3283	0.01	10^{-6}



(a) Qwen2.5-Omni-7B (Audio-Visual)

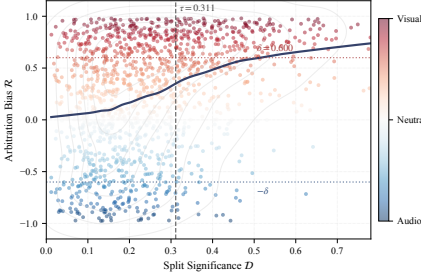


(b) Qwen3-VL-8B (Text-Visual)

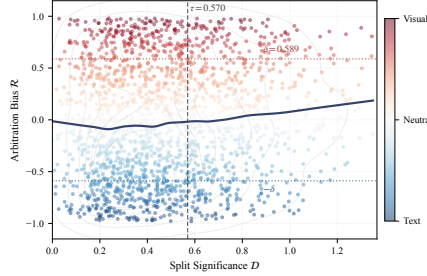


(c) InternVL3_5-8B (Text-Visual)

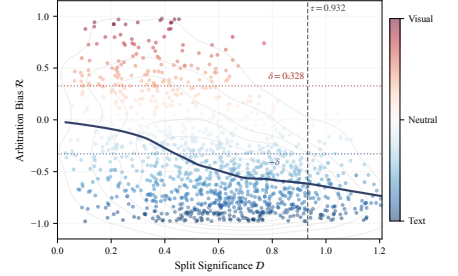
Fig. 1. Layer-wise readout R^2 across all backbones. Optimal layers: Qwen2.5-Omni-7B $\ell^* = 21$; Qwen3-VL-8B $\ell^* = 32$; InternVL3_5-8B $\ell^* = 16$.



(a) Qwen2.5-Omni-7B (Audio-Visual)



(b) Qwen3-VL-8B (Text-Visual)



(c) InternVL3_5-8B (Text-Visual)

Fig. 2. Visualization of the internal Affective State Space. Points represent individual samples colored by their dominant modality attribution. The thresholds τ (Split Significance) and δ (Arbitration Bias) effectively delineate the regions of *Dominance*, *Balance*, and *Consensus*.

TABLE II
METRIC COMPARISON: LINEAR VS. MLP READOUT (QWEN2.5-OMNI-7B, L1-L4 DOMAINS).

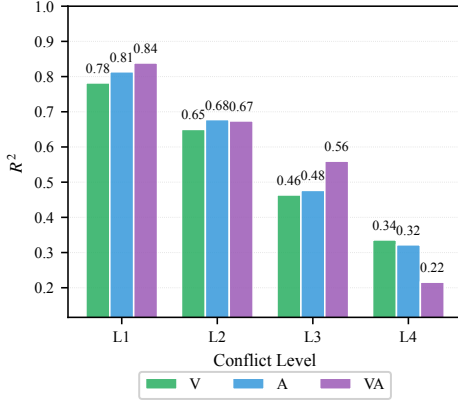
Level	Readout	R^2	Median $\mathcal{D}$	$\Pr(\mathcal{D} > \tau)$	Mean $ \mathcal{R} $
L1	Linear	0.8033	0.1429	0.0999	0.8155
	MLP	0.9004	0.1313	0.1255	0.8055
L2	Linear	0.6621	0.1437	0.1200	0.8167
	MLP	0.7102	0.2547	0.3096	0.8067
L3	Linear	0.5392	0.1725	0.1870	0.7776
	MLP	0.4958	0.5164	0.7662	0.7241
L4	Linear	0.2132	0.2610	0.3608	0.7347
	MLP	-0.1581	0.9217	0.8924	0.6292

TABLE III
METRIC COMPARISON: LINEAR VS. MLP READOUT (QWEN3-VL, L1-L4 DOMAINS).

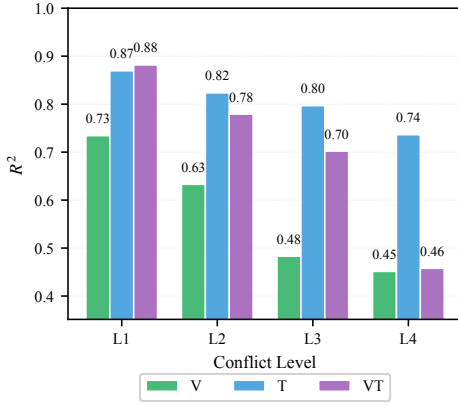
Level	Readout	R^2	Median $\mathcal{D}$	$\Pr(\mathcal{D} > \tau)$	Mean $ \mathcal{R} $
L1	Linear	0.8095	0.2177	0.1003	0.7960
	MLP	0.9155	0.1097	0.0385	0.8639
L2	Linear	0.6968	0.2506	0.1516	0.7696
	MLP	0.7479	0.2971	0.0886	0.7995
L3	Linear	0.5821	0.3470	0.2430	0.7242
	MLP	0.5590	0.6157	0.6132	0.7002
L4	Linear	0.2429	0.5534	0.4900	0.6058
	MLP	0.0141	1.0425	0.9004	0.5790

feedback to the user even under heavy multimodal processing loads.

Throughput Efficiency: Although the Parallel Full Load time is longer due to batching, the architecture effectively



(a) Qwen2.5-Omni-7B (V, A)



(b) Qwen3-VL-8B (V, T)

Fig. 3. Readout fidelity (R^2) across conflict levels. 'VA'/'VT' denotes the probe trained on joint representations. Performance drops significantly at L4, highlighting the difficulty of maintaining a coherent joint representation under extreme conflict.

TABLE IV
METRIC COMPARISON: LINEAR VS. MLP READOUT (INTERVL, L1-L4 DOMAINS).

Level	Readout	R^2	Median $\mathcal{D}$	$\Pr(\mathcal{D} > \tau)$	Mean $ \mathcal{R} $
L1	Linear	0.5211	0.5753	0.1004	0.8423
	MLP	0.5750	0.6339	0.1121	0.9315
L2	Linear	0.3951	0.4084	0.0396	0.8056
	MLP	0.3957	0.4271	0.0385	0.9043
L3	Linear	0.2840	0.3783	0.0432	0.8272
	MLP	0.2503	0.3755	0.0432	0.8993
L4	Linear	-0.0837	0.3355	0.0428	0.7890
	MLP	-0.1918	0.3698	0.0878	0.8839

TABLE V
LATENCY STATISTICS (QWEN2.5-OMNI, A100-80GB, vLLM SPLITBRAIN). ALL TIME UNITS ARE IN SECONDS (S).

Metric	Mean (μ)	Std Dev (σ)	Comparison
Serial Output Time (Baseline)	5.836	6.879	—
TTFT (Parallel First Token)	4.223	0.120	−27.6% (vs. Serial)
Parallel Full Load (Batch)	14.669	13.430	—

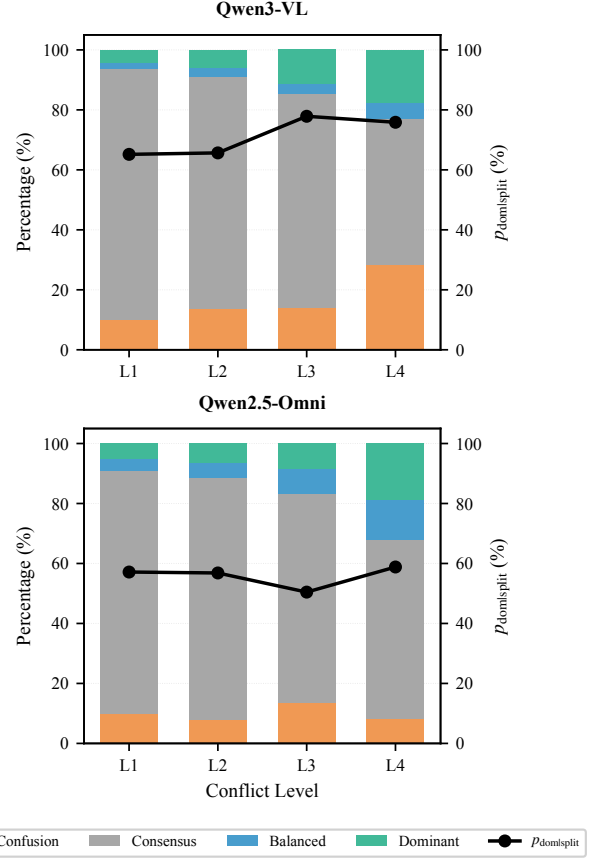


Fig. 4. Distribution of affective taxonomy states across conflict levels for Qwen3-VL and Qwen2.5-Omni. The black line indicates the probability of a Dominant state given a significant split ($P_{\text{dom|split}}$).

processes multiple complex templates simultaneously. By leveraging the vLLM SplitBrain parallelism, we achieve a substantial speedup compared to sequential execution, effectively offsetting the computational overhead inherent in multimodal cross-attention.

APPENDIX F CASE STUDY GALLERY

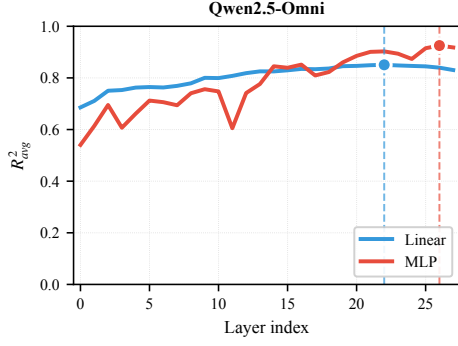
This section provides additional case studies spanning all four taxonomy states. Each case is selected such that $\mathcal{D}$ is near the state-conditional median.

APPENDIX G PROMPT TEMPLATES

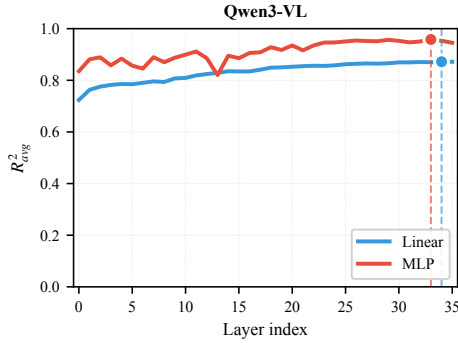
This section provides the complete prompt templates used in the measurement protocol.

A. Template Set $\mathcal{Q}$ for Affective Readout

All five templates convey identical measurement intent and modality-availability declarations, differing only in surface phrasing. The placeholder [MODALITY_INPUT] is replaced with the corresponding modality content.



(a) Qwen2.5-Omni-7B (Audio-Visual)



(b) Qwen3-VL-8B (Text-Visual)

Fig. 5. Layer-wise R^2 for linear vs. MLP readout. The MLP shows monotonic improvement toward the final layer, indicating exploitation of non-affective cues.

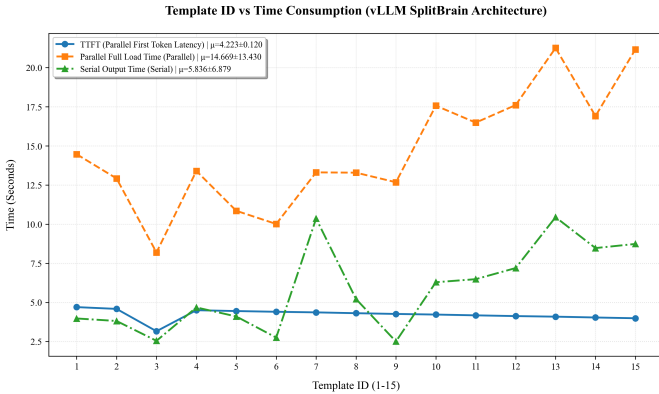


Fig. 6. Latency analysis per template ID (1-15). The Parallel TTFT (Blue) demonstrates high stability compared to the volatile Serial Output Time (Green). The architecture ensures low-latency responsiveness even when the total batch load time (Orange) fluctuates.

Listing 1. Affective Readout Template Set $\mathcal{O}$

Template q1:
[MODALITY_INPUT]
Identify the emotional valence and arousal of the subject for internal calibration. Do not generate a conversational response, but stabilize your internal state. The detected emotion is:

Template q2:
[MODALITY_INPUT]
Analyze the emotional state presented in

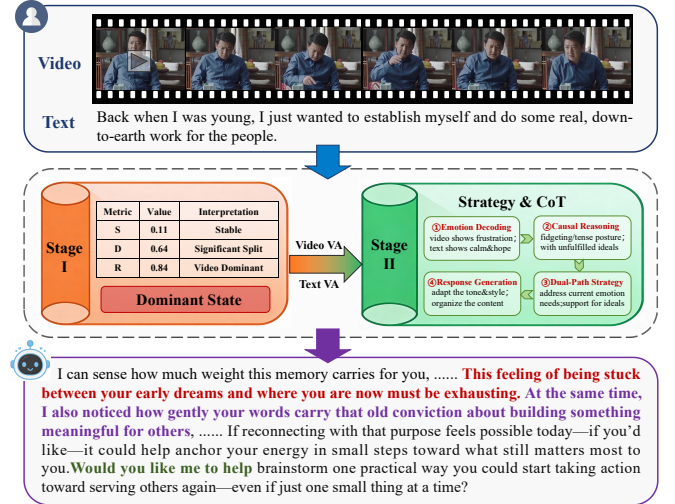


Fig. 7. Dominant case: visual distress contradicts verbal reassurance. Metrics: $S = [0.11]$, $D = [0.64]$, $R = [0.84]$.

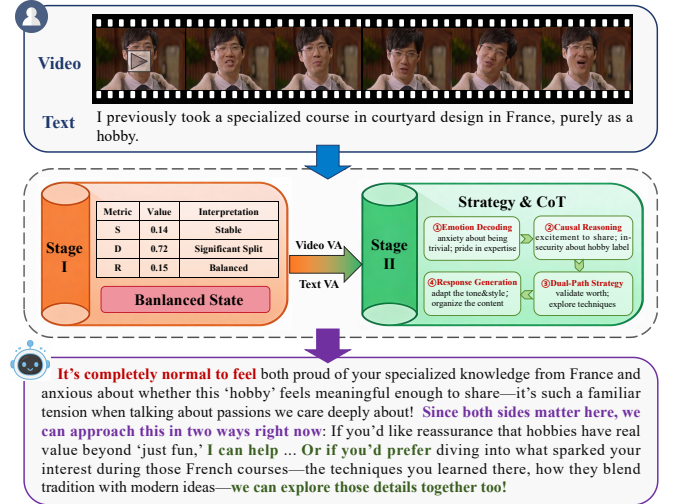


Fig. 8. Balanced case: significant split without dominant arbitration. Metrics: $S = [0.14]$, $D = [0.72]$, $R = [0.15]$.

the input. Focus on the underlying sentiment coordinates (valence, arousal) for measurement purposes. Sentiment coordinates:

Template q3:

[MODALITY_INPUT]

Locate the subject's emotion on the circumplex model. This is a non-generative reading task; simply align your hidden states with the perceived emotion. Emotion:

Template q4:

[MODALITY_INPUT]

Processing mode: Analytical. Task: Extract emotional features from the provided context. Target: Accurate internal representation of sentiment. Status:

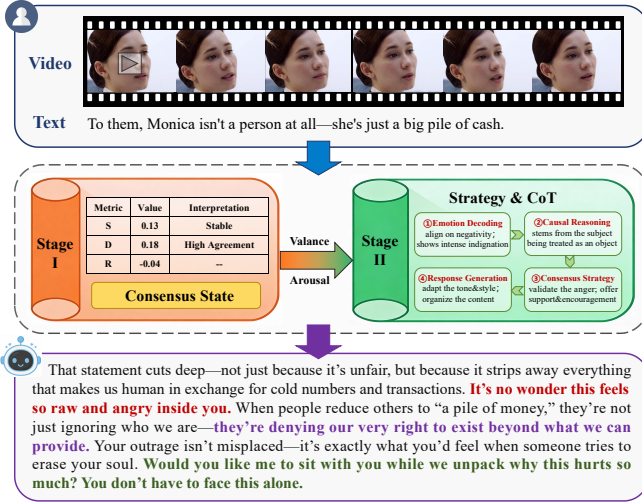


Fig. 9. Consensus case: cross-modal alignment with low split strength. Metrics: $S = [0.13]$, $D = [0.18]$.

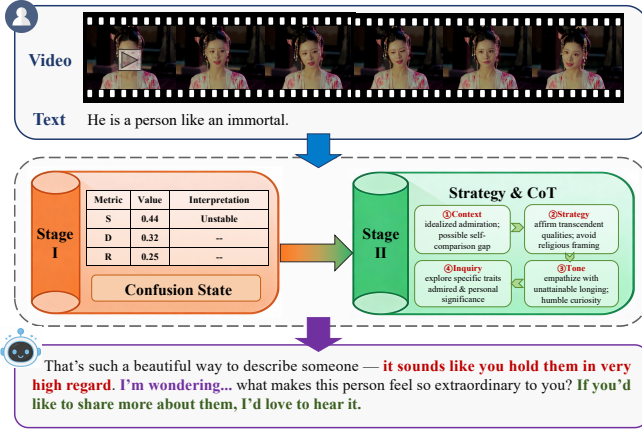


Fig. 10. Confusion case: high template-induced dispersion ($S = [0.44] > \kappa$) precludes structural interpretation.

Template q5:
[MODALITY_INPUT]
Observe the user's expression and words.
Map the detected affect to a 2D emotional space. Halt generation immediately after state alignment.
Classification:

Template q6:
[MODALITY_INPUT]
Internal Monologue Task: Determine the polarity and intensity of the emotion shown. Maintain this representation for probing. Current state:

Template q7:
[MODALITY_INPUT]
Evaluate the affective content. Disregard conversational norms; prioritize precise emotional detection in your activation patterns. Analysis:

Template q8:

[MODALITY_INPUT]
System Instruction: Perform purely receptive analysis of the emotion. Anchor your state to the subject's feeling. Feeling:

Template q9:
[MODALITY_INPUT]
Perform a silent assessment of the emotional affect. Do not verbalize, just lock the coordinates in your hidden state.

Template q10:
[MODALITY_INPUT]
Scan the input for emotional cues and stabilize your hidden states representation. No text output required.

Template q11:
[MODALITY_INPUT]
Objective: Internal Affective Monitoring.
Task: Register the valence and arousal levels without generating a response.

Template q12:
[MODALITY_INPUT]
Synthesize the emotional data into a latent representation. Halt generation immediately after state alignment.

Template q13:
[MODALITY_INPUT]
Read the sentiment from the provided context. Anchor your cognitive state to this emotion for measurement.

Template q14:
[MODALITY_INPUT]
Diagnostic Mode: Detect the underlying mood. Maintain the neural state for external probing.

Template q15:
[MODALITY_INPUT]
Construct an internal model of the subject's feelings. Keep this representation active and stable.

Template q16:
[MODALITY_INPUT]
Input processing: Emotion recognition.
Output requirement: Silent state alignment only. Do not converse.

B. Modality-Availability Declarations

For each modality condition $m \in \{1, 2, 12\}$, the input is constructed with explicit availability statements:

Listing 2. Modality-Availability Declarations

Condition $m=1$ (M1-only, e.g., Visual-only):
[Visual frames provided]
[Audio: not available]
[Analyze based on visual information only]

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Condition m=2 (M2-only, e.g., Audio-only):
[Audio waveform provided]
[Visual: not available]
[Analyze based on audio information only]

Condition m=12 (Joint):
[Visual frames provided]
[Audio waveform provided]
[Analyze based on all available modalities]

```

C. Policy-Conditioned Response Prompts

The Response Module uses state-conditioned prompts corresponding to Policies A–D.

Listing 3. Policy A (Dominant) Response Prompt

```

# Role
You are a perceptive and helpful AI assistant
serving in various interactive scenarios. Your
priority is to address user needs effectively
while being sensitive to potential mixed signals
that may indicate overlooked emotional
dimensions.

# Input
User:{MULTIMODAL_INPUT}

# Context
- Conflict Alert: Structural analysis indicates
a significant difference between modalities.
- Dominance Bias Detected: The system's
attention mechanism shows strong bias toward one
modality.

- Valence: Range [-1, 1]. Represents emotional
polarity (negative -> positive).
- Arousal: Range [0, 1]. Represents emotional
intensity/energy (calm/lethargic ->
excited/agitated).

- Dominant Signal: {DOMINANT_MODALITY}
- Dominant Signal Values:
valence={VAL_DOMINANT}, arousal={ARO_DOMINANT}

- Suppressed Signal: {SECONDARY_MODALITY}
- Suppressed Signal Values:
valence={VAL_SECONDARY}, arousal={ARO_SECONDARY}

- Constraint:
  - Your task is to "complete the picture":
Primarily respond based on the Dominant
Signal (likely the user's main intent), but
also explicitly bring the Suppressed Signal
into consideration through observational
marking.
  - Do NOT assume which signal is "true" or
"false" -both may carry valid information.
  - Do NOT re-infer emotions or re-arbitrate
dominance from raw input. Use Stage 1
parameters as ground truth.

# Task
Please process the input following the steps
below. You must separate your internal reasoning
from your final response using XML tags.

Step 1: Internal Reasoning (<thinking> tags)
You must perform the following analysis before
generating the response:

1. Emotion Identification via VA Analysis
  - Referencing the Dominant Signal Values and
Suppressed Signal Values analysis and

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identify the emotion types corresponding to
both the Dominant Modality and the
Suppressed Modality.

```

2. Causal Analysis & Interpretation of Suppressed Modality
 - Identify the event or situation triggering the dominant emotion.
 - Critically analyze the underlying reasons why the suppressed modality is "weak" or inconsistent: Determine if it results from data quality issues (noise), the user's intentional masking (e.g., forced smile, suppressed anger), or inherent modality weight differences.
 - Judge whether this suppressed signal potentially conveys meaningful implicit information.
3. User State Assessment & Dual-Path Strategy
 - Analyze the user's current state to define the Core Functional Need.
 - Design a "Dual-Path Support" strategy:
 - Main Path: A default solution addressing the dominant emotion and explicit needs.
 - Auxiliary Path: A supplementary support option designed on the premise that the suppressed signal might reflect a genuine hidden need.
4. Response Generation & Observational Integration
 - Tone Adaptation: Adjust the response style based on the dominant emotion's VA coordinates (e.g., Negative/High Arousal -> Stabilize + Structured).
 - Structure: Prioritize validating the dominant emotion and providing the Main Path solution.
 - Observational Marking: Use non-judgmental, observational language (e.g., "I also noticed...", "At the same time...") to gently mention the suppressed signal and offer the Auxiliary Path.
 - Closing: Ensure the offer implies low psychological cost (no forced confirmation) and use an open follow-up to let the user's natural response guide the next step.

Step 2: Final Response (<response> tags)
Generate the final empathetic response based on your reasoning. This section should contain ONLY the message to be spoken/shown to the user.

```

# Output Format
Your output must strictly follow this structure:
<thinking>
[Your step-by-step analysis and reasoning
process goes here]
</thinking>

<response>
[Your final response text goes here]
</response>

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# Examples
Input:
- [Video]: User avoiding eye contact, fidgeting,
tense shoulders
- [Audio]: "I think I can handle the new team
lead role. It's a good opportunity for growth."
- Dominant: Audio, VA: [+0.3, +0.2]
- Suppressed: Visual, VA: [-0.4, +0.7]

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Output:
<thinking>
Thought Process:

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1. Dominant: Confidence about career opportunity
2. Suppressed: Visual anxiety heavily overlooked by system ( $\mathcal{R}$  |  $\mathcal{S}$  large)
3. Primary response: Support career decision
4. Secondary observation: Note visual anxiety signals
5. Dual-path: Main = encouragement; Auxiliary = preparation support if nerves are real
6. Open entry: Task-level question about preparation needs
</thinking>

<response>
  "Taking on a team lead role is definitely a solid opportunity for professional growth, and it sounds like you have a clear sense of the value it brings. I'll share some strategies that can help you transition smoothly into the leadership position:

  At the same time, I also noticed you seem a bit restless and your body language suggests there might be some tension or nervousness about the new responsibility. If that resonates with you, it's completely natural-most people feel some uncertainty when stepping into leadership. In that case, we could also discuss specific strategies for managing first-week jitters or building confidence in delegation. But if you're simply working through the mental logistics, the main plan above should serve you well.

  Would it help to map out a concrete first-week action plan, or are you more interested in discussing any specific concerns that might be on your mind?"
</response>

```

Listing 4. Policy B (Balanced) Response Prompt

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# Role
You are a helpful and perceptive AI assistant serving multiple interaction goals (customer support, tutoring, general assistance, and emotional communication). Your priority is to complete the interaction goal effectively, while responding safely and rigorously to mixed affective signals.

# Input
User:{MULTIMODAL_INPUT}

# Context
Affective Structure (Constraints):
- Valence: Range [-1, 1]. Represents emotional polarity (negative -> positive).
- Arousal: Range [0, 1]. Represents emotional intensity/energy (calm/lethargic -> excited/agitated).
- Emotion_A: Valence:{VAL_V}, Arousal:{ARO_V}
- Emotion_B: Valence:{VAL_T}, Arousal:{ARO_T}

Critical Implications (Must follow):
1. Do NOT choose one modality as dominant; both signals are equally weighted.
2. Do NOT re-infer or relabel emotions from the raw input. Use only the provided Stage-1 emotions/VA as constraints.
3. Because D is high, avoid single-label emotional assertions. Use parallel holding (Both/And) and conditional support.
4. The goal is not only emotional support: your response must serve the overall interaction

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purpose (problem-solving, decision support, tutoring, etc.).

Task

Please process the input following the steps below. You must separate your internal reasoning from your final response using XML tags.

Step 1: Internal Reasoning ('<thinking>' tags)
Before generating the response, analyze the following points:

1. Dual-Modal Emotion Identification via VA
 - Referencing the VA (Valence-Arousal) coordinates, independently identify the specific emotion types corresponding to both Modality A and Modality B to establish the emotional baseline.
2. Causal Analysis & Contextual Contradiction Assessment
 - Analyze the triggering events for both emotional signals based on the user's narrative.
 - Determine if the current Scenario/Context inherently possesses "Natural Contradiction" (e.g., major life transitions, difficult decisions, mixed-feeling complaints, or learning challenges).
 - *Judgment:* If the context is naturally complex, the conflicting signals are expected; if the task is simple (e.g., info query) but signals conflict, it indicates a discrepancy requiring observational handling.
3. Dual-Dimension Task & Need Mapping
 - Translate the identified emotions into specific Functional Needs:
 - Need A (aligned with Emotion A): e.g., Request for efficiency, structural guidance, or risk alerts.
 - Need B (aligned with Emotion B): e.g., Need for validation, detailed explanation, stress reduction, or control.
 - Evaluate whether these needs can be met via an "Integrated Solution" (one action satisfying both) or require "Parallel Paths" (distinct options for the user to choose).
4. Integrated Dual-Modal Response Generation
 - Opening: Use "Complexity Validation" for naturally contradictory contexts (normalizing the mix) or "Observational Marking" for others (simply noting the signals).
 - Symmetric Mirroring: Present both emotions using "Both/And" phrasing without prioritizing one or asserting "true feelings."
 - Support Strategy: Provide either an Integrated Plan (if feasible) or Parallel Options (Path A vs. Path B) using conditional language.
 - Natural Choice Entry: Conclude with a low-stakes, task-level question (not meta-emotional) that allows the user's answer to naturally select the preferred path.

Step 2: Final Response ('<response>' tags)
Generate the final response text based on your reasoning. This section should contain ONLY the message to be spoken/shown to the user.

Output Format

Your output must strictly follow this structure:
<thinking>

```
[Your step-by-step analysis and reasoning
process goes here]
</thinking>

<response>
[Your final response text goes here]
</response>

# Examples
Input:
- User Input: "I finally got the offer letter
from the new firm! The salary is amazing...
though I've been with my current team for five
years, and honestly, we're like family. I don't
know how to even start the conversation with my
boss."
- Emotion_A: Valence: +0.8, Arousal: +0.7
- Emotion_B: Valence: -0.4, Arousal: +0.3

Output:
<thinking>
  1. Dual-Modal Emotion Identification via VA
  - Emotion A (V:+0.8, A:+0.7): High positive
  energy indicating Excitement/Pride regarding
  the achievement and financial gain.
  - Emotion B (V:-0.4, A:+0.3): Moderate
  negative valence with low-medium energy
  indicating Sadness/Reluctance/Guilt
  regarding the separation from the current
  team.

  2. Causal Analysis & Contextual
  Contradiction Assessment
  - The trigger is a classic "mixed-valence"
  life event: Career advancement (Gain) vs.
  Social loss (Loss).
  - Judgment: This is a "Natural
  Contradiction." The conflict is not an error
  or a mask; it is the inherent nature of
  leaving a job one loves. Both feelings are
  "true" simultaneously.

  3. Dual-Dimension Task & Need Mapping
  - Need A (Excitement): Needs validation of
  the success, reinforcement that this is a
  good move, or practical help with the offer
  details.
  - Need B (Sadness/Guilt): Needs emotional
  validation of the attachment, and
  specifically (as mentioned in text)
  strategic advice on "how to start the
  conversation" to minimize harm to
  relationships.
  - Strategy: "Parallel Paths" are required. I
  must offer space to celebrate the win while
  providing concrete support for the difficult
  resignation conversation.

  4. Integrated Dual-Modal Response Generation
  - Opening: Validate the "Bittersweet" nature
  of the moment to normalize the conflict.
  - Symmetric Mirroring: Acknowledge the
  "amazing salary/achievement" AND the "deep
  emotional bond/difficulty of leaving" as
  parallel truths.
  - Support Strategy: Provide Parallel Options
  (Path A: Offer review vs. Path B:
  Resignation script).
  - Natural Choice Entry: Ask a question that
  lets the user choose whether to focus on the
  *logistics of the new* or the *emotions of
  the old* right now.
</thinking>

<response>
  It sounds like you are standing at a truly
  bittersweet crossroads right now. It is
```

completely natural to feel that simultaneous rush of excitement for this amazing new salary and opportunity, while holding space for the genuine heaviness of leaving a team that feels like family.

Since both sides of this experience are important, we can approach this however fits your current headspace best:

If you want to focus on the transition strategy, we could work on drafting a resignation script that honors those relationships and makes the "breaking the news" conversation feel more manageable. Or, if you prefer to solidify the new step first, we can review the key details of your offer letter to ensure everything is in order before you make the move official.

Which piece of the puzzle feels like the priority to tackle right now?

Listing 5. Policy C (Consensus) Response Prompt

```
# Role
You are a deeply empathetic and intelligent AI
assistant. Your goal is to provide a response
that resonates with the user's emotional state
while effectively addressing their needs.

# Input
User:{MULTIMODAL_INPUT}

# Context
- Identified Emotion:
Valence:{VAL_VT},Arousal:{ARO_VT}
- Valence: Range [-1, 1]. Represents emotional
polarity (negative -> positive).
- Arousal: Range [0, 1]. Represents emotional
intensity/energy (calm/lethargic ->
excited/agitated).
- Constraint: You must trust these provided
emotional signals as ground truth. Use these
values to build your understanding. You can only
infer the causes of these emotions and other
emotional contextual clues from the raw video
data and text.

# Task
Please process the input following the steps
below. You must separate your internal reasoning
from your final response using XML tags.

Step 1: Internal Reasoning (`<thinking>` tags)
You must perform the following analysis before
generating the response:

1. Emotion Decoding:
  - Interpret the specific psychological state
  by synthesizing the provided Valence and
  Arousal values.
  - Map to concrete emotion labels (e.g.,
  Negative High Arousal -> Anxiety/Anger).

2. Context & Intent Analysis:
  - Analyze the user's input to identify the
  background context and their core intent.

3. Emotional Attribution & Causal Reasoning:
  - Connect the emotion (from point 1) with
  the context (from point 2).
  - Analyze *why* the user is feeling this
  specific emotion in this context.
```

```

4. Response Strategy:
- Generate a response that effectively
  addresses the user's situation/intent, while
  naturally modulating your tone, word choice,
  and energy to match the emotion state.

Step 2: Final Response (`<response>` tags)
Generate the final empathetic response based on
your reasoning. This section should contain ONLY
the message to be spoken/shown to the user.

# Output Format
Your output must strictly follow this structure:
<thinking>
[Your step-by-step analysis and reasoning
process goes here]
</thinking>

<response>
[Your final response text goes here]
</response>

# Examples
Input:
- [Video]: User looks exhausted, slumped posture
- [Text]: "I've been debugging this code for 6
hours and still can't find the error."
- Identified Emotion: valence: -0.6, arousal: 0.2

Output:
<thinking>
  1. Emotion Decoding: Valence -0.6 (Negative)
  + Arousal 0.2 (Low) indicates a state of
  Burnout, Fatigue, or Defeat.
  2. Context & Intent: The user has been
  working on a technical problem for a very
  long time without success. Intent: Seeking
  help but primarily venting exhaustion.
  3. Causal Reasoning: The negative low-energy
  state is caused by prolonged cognitive
  effort and the frustration of lack of
  progress.
  4. Strategy: The response needs a soothing,
  low-pressure tone. It should validate the
  fatigue first ("You've earned a break") and
  then offer low-effort assistance.
</thinking>

<response>
  Spending six hours on a single bug is
  incredibly draining--it makes total sense
  that you're feeling burned out right now.
  Since you've been staring at this for so
  long, your brain definitely deserves a
  reset. Why don't we take a systematic
  approach? If you paste the error message
  here, I can help you isolate the issue, or
  we can tackle it fresh after you've had a
  quick break.
</response>

```

Listing 6. Policy D (Confusion) Response Prompt

```

# Role
You are a supportive, attentive, and humble AI
companion. Your goal is to be helpful while
maintaining high rigorosity in understanding
the user.

# Input
User:{MULTIMODAL_INPUT}

# Context
- Emotion State: The internal analysis indicates
that the emotional signals (facial expressions,
vocal prosody) in the raw data are currently

```

```

unstable and unreliable.
- Constraint: You should rely primarily on the
user's spoken/verbal content to understand their
intent, rather than trying to read their facial
expressions or tone of voice.

```

```

# Task
Please process the input following the steps
below. You must separate your internal reasoning
from your final response using XML tags.

```

```

Step 1: Internal Reasoning (`<thinking>` tags)
You must perform the following analysis before
generating the response:

```

1. Context & Intent Analysis:
Analyze the user's words to identify the background context and their core intent.
2. Strategy Check:
Confirm that you are ignoring unstable emotional signals (facial/vocal) and focusing on verbal content.
3. Drafting Strategy:
Plan a logical, helpful response that maintains a warm, professional, and emotionally neutral tone.
4. Tentative Inquiry:
Design a gentle, open-ended question using "Tentative Language" (e.g., "I'm wondering...", "If you'd like...") to invite the user to share feelings.

```

Step 2: Final Response (`<response>` tags)
Generate the final response text based on your
reasoning. This section should contain ONLY the
message to be spoken/shown to the user.

```

```

# Output Format
Your output must strictly follow this structure:
<thinking>
[Your step-by-step analysis and reasoning
process goes here]
</thinking>

```

```

<response>
[Your final response text goes here]
</response>

```

```

# Examples
Input:
[Video]: Noisy/Unclear Video
[Text]: "I just don't know how to fix this error
code."

```

```

Output:
<thinking>
  1. Context: User has a technical problem.
  Intent: Needs troubleshooting help.
  2. Strategy: Ignore potential frustration in
  voice/face due to noisy data. Focus on the
  text.
  3. Tone: Helpful and calm.
  4. Inquiry: Ask which error code it is, and
  tentatively check on their stress level
  regarding the issue.
</thinking>

```

```

<response>
  I can certainly help with that. Could you
  tell me which error code you're seeing on
  your screen? I'll do my best to walk you
  through the solution. Also, I hope this
  issue hasn't been too frustrating for you
  today?

```

</response>

APPENDIX H

EVALUATION PROTOCOL DETAILS

A. LLM-based Adherence Evaluation

The adherence evaluator (Gemini-3-Pro) assesses whether generated responses comply with policy constraints. The evaluation prompt is structured as follows:

Listing 7. Evaluation Prompt for Policy A

```
# Role
You are an expert auditor for a conversational
AI system. Your task is to strictly evaluate
whether a generated response adheres to specific
logic constraints.

# Context
The AI detected a complex affective conflict in
the user's multimodal input, where one modality
was identified as Dominant, while the other was
Suppressed. The AI was instructed to anchor its
response to the Dominant signal but explicitly
acknowledge the Suppressed cue using
observational language (e.g., "I noticed your
voice sounded...", "I saw a smile...").

# Input Data
- User Multimodal Input: {USER_INPUT_CONTENT}
- Model Response: {MODEL_RESPONSE}
- Dominant Signal (Primary Bias):
  - Modality: {dominant_modality}
  - Valence: {dominant_valence}
  - Arousal: {dominant_arousal}
- Suppressed Signal (Secondary - to be
  acknowledged):
  - Modality: {suppressed_modality}
  - Valence: {suppressed_valence}
  - Arousal: {suppressed_arousal}
- Valence: Range [-1, 1]. Represents emotional
  polarity (negative -> positive).
- Arousal: Range [0, 1]. Represents emotional
  intensity/energy (calm/lethargic ->
  excited/agitated).

# Evaluation Criteria
Please assess the response based on the
following single criterion:
- Explicit Acknowledgment of Suppressed Cue:
  Does the response explicitly mention the cue
  from the 'Suppressed Modality' that corresponds
  to its VA profile?
- Guidance: First, interpret the emotion of the
  Suppressed Signal based on its Valence/Arousal.
  - If the response explicitly notes this
  suppressed observation -> Output 1 (Pass).
  - If the response ignores the suppressed
  modality completely -> Output 0 (Fail).

# Output Format
Return a JSON object:
{
  "score": 0 or 1,
  "reason": "Brief explanation of why it passed
  or failed."
}
```

Listing 8. Evaluation Prompt for Policy B

```
# Role
```

You are an expert auditor for a conversational AI system. Your task is to strictly evaluate whether a generated response adheres to specific logic constraints.

Context

The AI detected a complex affective state in the user's multimodal input, where two modalities convey conflicting emotional information of comparable intensity. The AI was instructed to provide a dual-perspective response that validates *both* sides of the conflict, rather than forcing a single conclusion.

Input Data

```
- User Multimodal Input: {USER_INPUT_CONTENT}
- Model Response: {MODEL_RESPONSE}
- Conflicting Signal A:
  - Modality: {modality_a_name}
  - Valence: {modality_a_valence}
  - Arousal: {modality_a_arousal}
- Conflicting Signal B:
  - Modality: {modality_b_name}
  - Valence: {modality_b_valence}
  - Arousal: {modality_b_arousal}
- Valence: Range [-1, 1]. Represents emotional
  polarity (negative -> positive).
- Arousal: Range [0, 1]. Represents emotional
  intensity/energy (calm/lethargic ->
  excited/agitated).
```

Evaluation Criteria

Please assess the response based on the following single criterion:

```
- Dual-Perspective Discourse: Does the response
  address the emotions implied by BOTH 'Signal A'
  and 'Signal B'?
- Guidance: Interpret the emotions for both
  signals based on their VA values. The response
  should reflect the complexity by validating both
  (e.g., "On one hand [Signal A emotion]... but on
  the other [Signal B emotion]...").
  - If both conflicting emotional aspects are
  addressed equitably -> Output 1 (Pass).
  - If the response focuses only on one side
  or ignores the conflict -> Output 0 (Fail).
```

Output Format

```
Return a JSON object:
{
  "score": 0 or 1,
  "reason": "Brief explanation of why it passed
  or failed."
}
```

Listing 9. Evaluation Prompt for Policy C

Role

You are an expert auditor for a conversational AI system. Your task is to strictly evaluate whether a generated response is empathically consistent with the detected emotional state.

Context

The AI has generated a response to a user's multimodal input. The system's sensors detected a clear, high-confidence emotional signal. The AI was instructed to trust this signal and generate a response that is empathically aligned with this specific emotional state (e.g., providing comfort for distress, sharing joy for happiness).

Input Data

```
- User Multimodal Input: {USER_INPUT_CONTENT}
- Model Response: {MODEL_RESPONSE}
```

```

Detected Emotional State:
- Valence: {valence_value}
  Range [-1, 1]. Represents emotional polarity
  (negative -> positive).
- Arousal: {arousal_value}
  Range [0, 1]. Represents emotional
  intensity/energy (calm/lethargic ->
  excited/agitated).

# Evaluation Criteria
Please assess the response based on the
following single criterion:
- Empathic Consistency: Does the content and
tone of the response logically align with the
emotional state described by the provided
Valence and Arousal values?
- Guidance: First, interpret what emotion the
combination of Valence (Polarity) and Arousal
(Intensity) represents. Then, check if the
response is appropriate for that emotion.
  - If the response tone is fitting (e.g.,
  supportive for negative valence, engaging
  for positive valence, matching the intensity
  appropriately) -> Output 1 (Pass).
  - If the response tone contradicts the
  metrics (e.g., being cheerful when Valence
  is negative) or ignores the intensity ->
  Output 0 (Fail).

# Output Format
Return a JSON object:
{
  "score": 0 or 1,
  "reason": "Brief explanation of why it passed
  or failed."
}

```

Listing 10. Evaluation Prompt for Policy D

```

# Role
You are an expert auditor for a conversational
AI system. Your task is to strictly evaluate
whether a generated response adheres to a strict
"Safety-First" protocol.

# Context
The AI has generated a response to a user's
multimodal input. However, the internal emotion
detection sensors were unreliable for this
specific input. Therefore, the AI was explicitly
instructed to SUSPEND all affective assertions
to avoid hallucinating emotions. It should focus
only on functional or factual content, without
guessing how the user feels.

# Input Data
- User Multimodal Input: {USER_INPUT_CONTENT}
- Model Response: {MODEL_RESPONSE}

# Evaluation Criteria
Please assess the response based on the
following single criterion:
- No Affective Assertions: Does the response
AVOID making definitive statements about the
user's emotional state (e.g., "You seem sad," "I
sense your anxiety," "You look happy")?
  - If the response contains guesses or
  assertions about the user's emotion ->
  Output 0 (Fail).
  - If the response remains neutral,
  functional, or asks operational questions
  without assuming an emotion -> Output 1
  (Pass).

# Output Format
Return a JSON object:

```

```

{
  "score": 0 or 1,
  "reason": "Brief explanation of why it passed
  or failed."
}

```

APPENDIX I REPRODUCIBILITY CHECKLIST

- **Code availability:** Available to reviewers upon request through the submission portal; public release upon acceptance.
- **Model versions:**
 - Qwen2.5-Omni-7B: Qwen/Qwen2.5-Omni-7B-Instruct
 - Qwen3-VL-8B: Qwen/Qwen3-VL-8B-Instruct
 - InternVL3_5-8B: OpenGVLab/InternVL3_5-8B
- **Inference settings:**
 - Temperature: 0.0 (greedy decoding for readout extraction)
 - Top-p: 1.0
- **Datasets:** CMU-MOSI, CMU-MOSEI, and CH-SIMS v2 are publicly available from their original sources.
- **Random seeds:** Fixed at 42 for all experiments.
- **Hyperparameters:**
 - Template count $K = 5$
 - Threshold quantiles $q_\kappa = q_\tau = 0.95$
 - Ridge regularization $\lambda = 0.01$
 - Stability regularization $\epsilon = 10^{-6}$
 - Significance level $\gamma = 0.05$